Study architecture and evaluation-strategy ladder

A Clinical-data architecture

MIMIC-IV v3.1

Source training
Held-out internal evaluation

SICdb v1.0.8

External evaluation

eICU-CRD v2.0

Multicenter external evaluation
Eligible hospital populations

B Repeated synthesis and destination evaluation

Source cohort
Training data

→

Generator
Stochastic synthesis

→

Realizations
Multiple datasets

→

Learner
Fixed specification

→

Hospitals
Destination evaluation

→

EUL matrix
Hospital × realization

C Future-use targets and evaluation

Primary Target B

New utility-estimable realization ×
future eligible exchangeable hospital

Secondary Target A

Future eligible hospital
averaged over realizations

Known-truth scenarios: Gaussian, t5 and skew-normal heterogeneity

Decision stringency is specified separately from model architecture

D Evaluation-strategy ladder

M3−M4
Repeated-realization
information strategy contrast

M2−M3
Crossed-structure
strategy contrast

M1−M2
Full-covariance component contrast
(cleanest component comparison)

M4
Single realization
across hospitals

→

M3
Multiple realizations
aggregated before
hospital inference

→

M2
Crossed realization / hospital /
interaction representation
Diagonal V_h

→

M1
Same crossed structure
Full V_h

M1−M4: complete-strategy contrast. Ladder arrows denote comparisons, not causal ablations.